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Significant Expansion of *Vicia pannonica* Genome Size Mediated by Amplification of a Single Type of Giant Retroelement

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ABSTRACT

Amplification and eventual elimination of dispersed repeats, especially those of the retroelement origin, account for most of the profound size variability observed among plant genomes. In most higher plants investigated so far, differential accumulation of various families of elements contributes to these differences. Here we report the identification of giant Ty3/*gypsy*-like retrotransposons from the legume plant *Vicia pannonica*, which alone make up ~38% of the genome of this species. These retrotransposons have structural features of the Ogre elements previously identified in the genomes of pea and Medicago. These features include extreme size (25 kb), the presence of an extra ORF upstream of the gag–pol region, and a putative intron dividing the prot and rt coding sequences. The Ogre elements are evenly dispersed on *V. pannonica* chromosomes except for terminal regions containing satellite repeats, their individual copies show extraordinary sequence similarity, and at least part of them are transcriptionally active, which suggests their recent amplification. Similar elements were also detected in several other Vicia species but in most cases in significantly lower numbers. However, there was no obvious correlation of the abundance of Ogre sequences with the genome size of these species.

NUCLEAR genomes of higher plants differ considerably in their size, ranging from 0.1 pg (98 Mbp) in *Fragaria viridis* to 89.5 pg (87,686 Mbp) in *Fritillaria davisii* (Bennett and Leitch 2004). Even closely related species belonging to the same genus can display 5- to 10-fold differences in their haploid genome size, as it is, for example, in Phalenopsis, Scilla, or Vicia (Bennett and Leitch 2005). First investigations of this phenomenon using DNA reassociation kinetics (Chooi 1971; Flavell *et al.* 1974) revealed that genome size variation is mainly caused by differences in the proportion of repetitive DNA sequences. This was later confirmed by finding many families of repetitive sequences from a number of species. Among these, satellite repeats and retroelements have the most significant impact on genome size. Satellite repeats are organized as long arrays of tandemly repeated units (monomers). Although the monomer sequences are usually only tens to hundreds of nucleotides long (Macas *et al.* 2002), they can be amplified up to millions of copies (Kato *et al.* 1984; Ingham *et al.* 1993; Irifune *et al.* 1995; Macas *et al.* 2000), making up to 20% of the genome (Ingham *et al.* 1993). However, in most plant species investigated so far, the majority of repetitive DNA is composed of various families of retroelements (reviewed in Kumar and Bennetzen 1999; Feschotte *et al.* 2002). This high proportion of retroelements within plant genomes is a consequence of their replicative (copy-and-paste) mode of transposition (retrotransposition), which generates a new copy of the element each time it is retrotransposed. Although the retroelements do not attain as high copy numbers as the satellite repeats, their impact on the genome size is more pronounced due to their considerable length, ranging from a few up to 14 kb (Hirochika *et al.* 1992; Martinez-Izquierdo *et al.* 1997; Neumann *et al.* 2005). The recent discovery of a new group of giant retrotransposons, named Ogre elements, showed that the upper length limit of retrotransposons could be even longer. These Ty3/*gypsy*-like elements identified in pea (*Pisum sativum*) and *Medicago truncatula* are up to 22 kb long, and they occur at ~10,000 copies in the pea genome, corresponding to at least 5% of its nuclear DNA (Neumann *et al.* 2003).

The genus Vicia (Fabaceae) includes >160 species differing considerably in their haploid nuclear DNA content (1.9–14.4 pg, corresponding to 1862–14,112 Mbp) (Bennett and Leitch 2005). Several studies revealed that there is a number of differentially amplified repeats of retroelement origin that significantly contribute to these differences (Pearce *et al.* 1996; Kumar *et al.* 1997; Nouzová *et al.* 2001; Hill *et al.* 2005). However, these studies described only partial retroelement sequences, which do not allow precise evaluation of the contribution that specific element families give to the evolution

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of the Vicia genome. In this work, we show that many of these partial sequences belong to a retrotransposon family closely related to the giant Ogre elements previously identified in pea and Medicago. We describe full-length Ogre-like elements isolated from the genome of *Vicia pannonica* and present data suggesting that significant expansion of the genome size in some Vicia species was caused by recent amplification of these elements.

MATERIALS AND METHODS

Plant material: Seeds of *V. pannonica* (cv. Dětěnická panonská) and other species used in this study were obtained from the seed bank of the Institute of Plant Genetics and Crop Plant Research, Gatersleben, Germany; Agritec Šumperk, Czech Republic; and Plant Breeding Station, Boršov, Czech Republic. Total genomic DNA was extracted from leaves as described by Dellaporta *et al.* (1983). All DNA concentration measurements were done using the PicoGreen dye (Molecular Probes, Eugene, OR) according to manufacturer's recommendations. The measurements were performed in microwell plates and PicoGreen fluorescence was evaluated using a fluoroimager (Typhoon 9410, Amersham, Buckinghamshire, UK). The whole set of genomic DNAs and control fragments used for dot blotting was measured simultaneously, using the same DNA concentration standards.

Cloning procedures and sequence analysis: Restriction fragments appearing as bands on ethidium-bromide-stained agarose gels of *V. pannonica* DNA digested with *Bgl*II or *Kpn*I were cut out of the gel, purified, and cloned into pBluescript II SK+ (Stratagene, La Jolla, CA) digested with the corresponding enzyme. The fragments were sequenced using the dideoxy-mediated chain-termination method (Sanger *et al.* 1977).

A cosmid library was prepared by partial digestion of high-molecular-weight genomic DNA of *V. pannonica* with *Mbo*I, followed by its dephosphorylation and cloning into *Bam*HI-digested vector SuperCos1 (Stratagene). The library was screened with the cloned *Bgl*II and *Kpn*I fragments as probes using AlkPhos direct hybridization and detection kit (Amersham). A clone selected for sequencing (VP-cosC6) was subcloned and the sequencing templates from individual subclones were prepared using the GeneJumper primer insertion kit (Invitrogen, San Diego). Sequence assembly and basic analysis was done with Staden Package software (Staden 1996). As the cosmid clone contained several highly similar elements, the sequence assembly of the whole clone was verified by its restriction analysis and by PCR with primers specific for its individual parts. Computer analysis of the resulting sequence was performed using the Dotter program (Sonnhammer and Durbin 1995), program tools implemented at the Biology WorkBench website (http://workbench.sdsc.edu/), and EMBOSS (Rice *et al.* 2000). Multiple-sequence comparisons were done using Clustal W (Thompson *et al.* 1994). BLAST and FASTA (Pearson and Lipman 1988; Altschul *et al.* 1997) were employed for homology searches and the RPS-Blast (Marchler-Bauer *et al.* 2003) was used to search for conserved protein domains. Phylogenetic analyses were done using Clustal W and a phylogenetic tree was reconstructed using the Phylojava client/server tool. Sequences of reverse transcriptase domains were taken mostly from the alignment ALIGN_000602 (Vicient *et al.* 2001). Sequences of tRNAs used for identification of the primer binding site (pbs) were obtained from the *Arabidopsis thaliana* tRNA database (Lowe and Eddy 1997). Splice-site analysis was performed at the NetGene2 server (Hebsgaard *et al.* 1996).

Copy-number estimation: To estimate the copy number of Ogre elements in the genomes of selected species, serial dilutions of their genomic DNAs were quantitatively dot blotted on Hybond N+ membrane (Amersham) together with fragments of the Ogre element as hybridization standards. The quantity of spotted genomic DNA corresponding to 50–10^5 copies of the haploid genomes (1C) was compared to 5×10^6–1×10^{10} copies of the hybridization standards. Dot blots were hybridized with the probes specific for different regions of the Ogre elements (LTR, ORF1, -2, and -3) prepared by PCR using VP-cosC6 DNA as a template and employing the following primers: VP1 5′-AAC TTT TAG TCA TTT ACT TTC AAT AAA CA-3′ and MT-pbs 5′-TCC CCA GTG AAG TCG CCA-3′; VP17 5′-TGG GAA GAA GAA ACA CCA AG-3′ and VP18 5′-CAT CTT CAT TTG ACG AGC AA-3′; VP19 5′-AAC GAG CTT CGT GGT ACA AT-3′ and VP20 5′-CTC GAG GAT TGT TGT GAC AG-3′; VP23 5′-CGA AGA GGA TGA AGA AGA GG-3′; and VP24 5′-TTT CTT GAC TGC ATC AGC AT-3′. Probe positions in the Ogre-VP1 element are shown in the Figure 2A. Probe labeling and hybridization were done using the AlkPhos kit following the manufacturer's instructions. Hybridization and washing temperatures were 65°. Probe detection was performed using CDP-*Star* substrate (Amersham) and the signals were captured on X-ray film. The signals were also captured and quantified using Typhoon 9410 scanner and ImageQuant TL software (Amersham).

Alternatively, estimation of Ogre copy numbers in the *V. pannonica* genome was based on the number of positive clones observed after screening a short-insert shotgun genomic library with the probes described above. The shotgun library was prepared from *V. pannonica* DNA subjected to sonication, mung bean nuclease treatment, and size fractionation on an agarose gel. The 600- to 800-bp fragments were purified from the gel, treated with polynucleotide kinase in the presence of ATP, and cloned into dephosphorylated *Sma*I-cut plasmid vector (pBluescript II SK+). Copy number (CN) was calculated for individual probes using the formula $CN = GS \times PG/L_p$, where GS is the genome size of *V. pannonica* (6.51×10^9 bp), PG is the proportion of the probe sequence in the genome, and L_p is the probe length. The PG-value was defined as $PG = L_h/L_t$, where L_h is the length of hybridizing sequences within the library and L_t is the sum of insert lengths of all screened clones. As it was supposed that many of positive clones did not hybridize over their whole length, the L_h-value could not be calculated by simply multiplying the number of positive clones by the average insert length. Instead, the L_h-value was calculated considering the theoretical frequency of clones hybridizing over their whole length (F_1), the theoretical frequency of clones hybridizing with only part of their sequences (F_2), the number of positively hybridizing clones (N_p), the average insert size ($L_i = 700$ bp), the minimum length of sequence capable of efficient hybridization ($L_{min} = 100$ bp), and the average length of partially hybridizing sequences [$L_{avg} = (L_i - L_{min} - 1)/2$], using the formula $L_h = N_p \times F_1 \times L_i + N_p \times F_2 \times L_{avg}$. The F_1- and F_2-values were calculated using the formulas $F_1 = (L_p - L_i)/((L_p - L_i) + 2 \times (L_i - L_{min} - 1))$ and $F_2 = 2 \times (L_i - L_{min} - 1)/((L_p - L_i) + 2 \times (L_i - L_{min} - 1))$.

RNA isolation and RT–PCR: The tissues used for RNA isolation (leaf, root, flower) were harvested from plants cultivated in pots under a 15 hr light/9 hr dark photoperiod at 22° and 18°, respectively. Total RNA was isolated using the total RNA isolation kit (Ambion, Austin, TX). All RNA samples were treated with RNase-free DNase (Invitrogen) to remove any contaminant DNA. Reverse transcription was carried out using the SuperScript II reverse transcription kit (Invitrogen) by a random priming method according to the manufacturer's recommendations, using 0.5 μg of template RNA. The RT–PCR reaction mix (25 μl) consisted of 1 × PCR buffer, 0.2 mM dNTPs, 0.2 μM primers, 1.5 mM $MgCl_2$, 1 unit of platinum *Taq*

polymerase (Invitrogen), and 2.5 ng of reversely transcribed RNA or an equal amount of reverse-transcriptase-untreated RNA as a negative control. The following primers were used in the RT–PCR experiments: VP17 and VP18; VP25 5′-ACG TTC TCT TTC ATC GAT GC-3′ and MT3 5′-CGG TAG TCA ACA CAC ATT CTG AC-3′. The reaction profile included 35 cycles of 30 sec at 94°, 50 sec at 55°, and 1–3 min at 72°, preceded by initial denaturation (3 min at 94°) and followed by a final extension step (10 min at 72°). Reaction products were resolved on agarose gel electrophoresis.

Fluorescence *in situ* hybridization: Fluorescence *in situ* hybridization (FISH) was performed on isolated chromosomes prepared as described by Gualberti *et al.* (1996) and centrifuged onto slides using a Hettich centrifuge equipped with cytospin chambers. The Ogre probe was derived from the longest *Eco*RV restriction fragment of the clone VP-cosC6 containing LTR, 5′-untranslated region (5′-UTR), ORF1, and ORF2 regions. The probe was labeled with biotin-16-dUTP (Boehringer Mannheim, Indianapolis) using random priming and detected using streptavidin-Alexa Fluor-568 (Molecular Probes) as described by Leitch *et al.* (1994). Treatment of slides before hybridization, composition of hybridization mix, and hybridization conditions of the Ogre probe were as described by Neumann *et al.* (2001). Following hybridization of the Ogre probe, the slides were dehydrated using the ethanol series, air dried, and used for the second round of hybridization with fluorescein-labeled oligonucleotide probe (5′-AAG ATT RTC TTG TGY TAT AST ACA TAA AAK TCA CGA AGT-3′) specific for satellite repeats VicTR-A (Macas *et al.* 2000), which produce labeling patterns allowing discrimination of all chromosome types within the *V. pannonica* karyotype (Navrátilová *et al.* 2003). Hybridization of the VicTR-A probe (0.5 ng/μl) was performed at 37° for 16 hr in the hybridization mix consisting of 2× SSC, 100 ng/μl sheared calf thymus DNA, and 0.125% SDS, and posthybridization washes were done in 2× SSC at 42° for 10 min. Chromosomes were counterstained with 4′,6-diamidino-2-phenylindole (DAPI) and examined using a Nikon Eclipse 600 microscope equipped with appropriate filter sets. The images were captured with a CCD camera and analyzed using Lucia software (Laboratory Imaging).

RESULTS

In our preliminary experiments aimed at comparative analysis of Vicia genomes, we noted that *V. pannonica* produces characteristic restriction patterns when its DNA is digested with restriction endonucleases and resolved on agarose gels. Contrary to the majority of species used for the experiments, which produced smears with a few distinct bands, these patterns consisted of a series of well-distinguished bands in most enzymes tested (Figure 1A). This observation suggests that there is a prominent and conserved DNA repeat making up a large portion of the *V. pannonica* genome. To isolate and characterize this repeat, we cloned four of the distinct bands from the DNA digested with *Bgl*II (850 and 1250 bp) or *Kpn*I (560 and 1000 bp). Sequence analysis of the cloned fragments revealed their similarity to different parts of the giant retroelement Ogre previously described in pea (Neumann *et al.* 2003). To isolate the corresponding full-length element from *V. pannonica*, a cosmid library was constructed and screened with the cloned fragments as probes. Thirty randomly picked clones hybridizing to all four probes were subjected to restriction analysis to check for the presence of fragments corresponding to the conserved genomic bands. These fragments were found in most of the clones, 13 of which (43%) produced bands corresponding to all four fragments used as probes. One such clone (VP-cosC6) was selected for complete sequencing and further analysis.

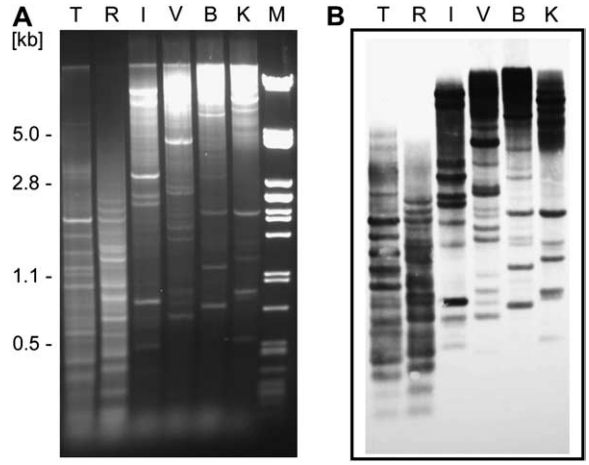


Figure 1.—*V. pannonica* Ogre sequences are highly conserved in restriction sites. (A) Genomic DNA of *V. pannonica* digested with *Taq*I (T), *Rsa*I (R), *Eco*RI (I), *Eco*RV (V), *Bgl*II (B), and *Kpn*I (K). λDNA digested with *Pst*I was used as a size marker (M). (B) Southern blot hybridization of the gel shown in A using the whole VP-cosC6 insert sequence as a probe.

Sequence characterization and transcriptional activity of Ogre elements in *V. pannonica*: The insert cloned in VP-cosC6 (GenBank accession no. AY936172) was >42 kb in length and entirely composed of Ogre sequences belonging to three different elements (Figure 2). There was one complete element, designated Ogre-VP1, which contained intact LTRs surrounding internal coding regions and was flanked by a 5-bp target-site duplication (5′-ATGCC/ATGCC-3′). This element was inserted into another Ogre element (Ogre-VP2), whose sequence was truncated at its left LTR due to cloning and at its right LTR due to insertion of the third element, Ogre-VP3. Most of the Ogre-VP3 sequence was lost through cloning except for a part of its left LTR. All three elements shared high mutual similarities. The overall similarity between Ogre-VP1 and Ogre-VP2 was 92.3%. The Ogre-VP3 LTR shared 92.1% similarity with the LTR sequence of Ogre-VP1 and 81.5% similarity with LTR of Ogre-VP2. Similarity of these elements to the partial sequences cloned as conserved *Bgl*II and *Kpn*I fragments was 86–99%.

The only full-length element, Ogre-VP1, was 25,049 bp long, which makes it the largest plant retroelement described so far. This extreme size was mainly due to exceptionally long LTRs, each of which spans 6438 bp. Sequence similarity between the LTRs was 99.9% (only

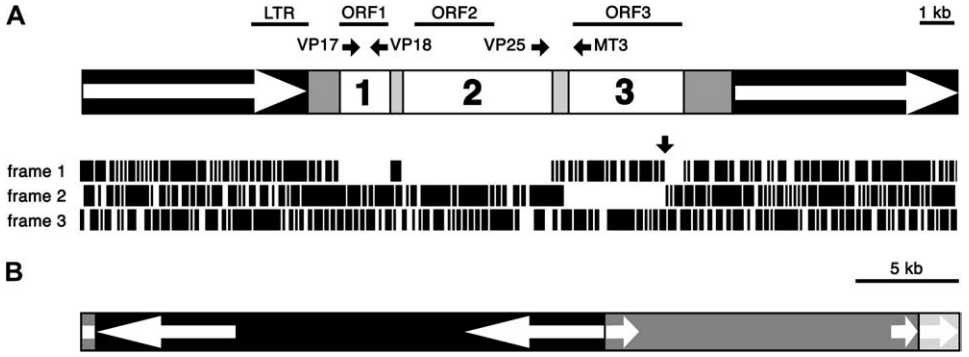


FIGURE 2.—Ogre elements identified in *V. pannonica*. (A) Structure of the full-length element Ogre-VP1. LTRs are depicted as open arrows in solid boxes. The open reading frames ORF1, ORF2, and ORF3 are represented by open boxes marked with corresponding numbers. Lightly shaded boxes indicate regions separating individual ORFs and darkly shaded boxes represent putative 5′- and 3′-UTRs. Lines above the scheme show positions of probes used for hybridizations. The horizontal solid arrows mark positions and directions (5′ → 3′) of primers used for RT–PCR experiments. Positions of stop codons in three reading frames of the Ogre-VP1 sequence are shown as vertical lines below the scheme. The vertical arrow indicates the frameshift in the ORF3. (B) Structure of the clone VP-cosC6 (GenBank accession no. AY936172) showing positions and orientation of Ogre-VP1 (solid), Ogre-VP2 (dark shading), and Ogre-VP3 (light shading) elements. Open arrows represent LTRs.

six nucleotide substitutions over their entire length), which indicates that Ogre-VP1 represents a very recent insertion. The overall arrangement of Ogre-VP1 coding and structural regions is identical to that of the Ogre elements identified in the *P. sativum* and *M. truncatula* genomes (NEUMANN *et al.* 2003) and further confirms its assignment into this group of Ty3/*gypsy*-like retrotransposons. Close relationships of Ogre sequences from all three species were also apparent from a phylogenetic analysis of the reverse transcriptase protein domains (supplementary Figure S1 at http://www.genetics.org/supplemental/). The coding region of Ogre-VP1 is divided into three reading frames (ORF1–3, Figure 2A) separated by short regions containing several stop codons in all three frames. These ORFs could be directly translated into protein sequences, with the exception of ORF3, containing a +1 frameshift due to an insertion of one nucleotide, which had to be removed to allow its conceptual translation.

Similarly to the pea elements, ORF1 of Ogre-VP1 encoded a protein with unknown function. This ORF was 1527 bp in length and was separated from ORF2 by a 237-bp-long noncoding region containing stop codons. The partial Ogre-VP2 sequence contained an ORF1 of the same length with 97% sequence similarity to ORF1 in Ogre-VP1. All of the protein domains typical for plant retroelements could be recognized within two ORFs: ORF2 and ORF3. Gag and protease (prot) domains were encoded by ORF2, whereas reverse transcriptase (rt), ribonuclease H (rh), and integrase (int) were encoded by ORF3 (Table 1). These ORFs were separated by a region of 317 bp, which contained multiple

TABLE 1

Conserved domains found in the putative polyprotein encoded by Ogre-VP1

CD[a]	*E*-value	Position[b]	Active-site motif	Note
gag (pfam03732)	$3e^{-09}$	264–350		The domain is followed by putative zinc-finger motif $CX_2HX_5HX_4C$ (513–527)[f]
prot	—	986–1054	DTG[c]	The domain was identified based on similarity to pea Ogre element (accession no. AY299398)
rt (pfam00078)	$7e^{-24}$	1624–1793	YVDD[d]	
rh (pfam00075)	$1e^{-08}$	2053–2173	DEDD[e]	
int (pfam00665)	$2e^{-21}$	2334–2488	DDE[g]	The domain is preceded by zinc-binding motif HHCC (2280–2314)[g]

[a] Domains typical for LTR retrotransposon polyproteins were identified by searching the conserved domain (CD) database using RPS-Blast (MARCHLER-BAUER *et al.* 2003) and by noting similarity to the putative protein encoded by the pea Ogre element identified in the clone Ps-cos16 (accession no. AY299398).

[b] The putative polyprotein sequence was deduced from ORF2 and ORF3, which were joined after the removal of predicted intron.

[c] SKALKA (1989).

[d] XIONG and EICKBUSH (1990); DING *et al.* (1998).

[e] DAVIES *et al.* (1991); MALIK and EICKBUSH (2001).

[f] BURD and DREYFUSS (1994).

[g] ASANTEAPPIAH and SKALKA (1997); MAIGNAN *et al.* (1998).

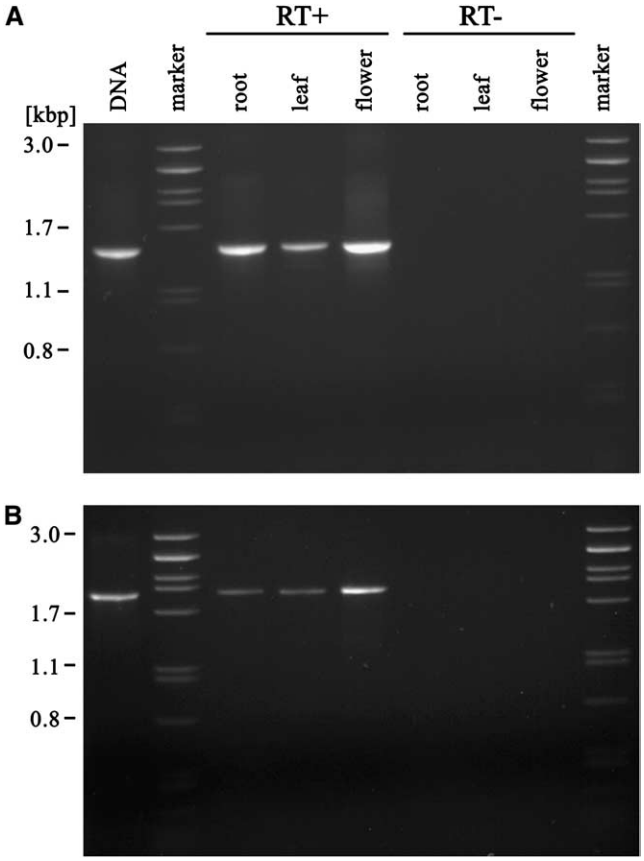


FIGURE 3.—Transcriptional activity of Ogre retrotransposons in *V. pannonica* analyzed by RT–PCR. The reactions were performed on total RNA isolated from roots, leaves, or flowers (RT+). Negative controls with omitted reverse transcriptase step were included to check for false-positive results (RT−). (A) Amplification of the ORF1 sequence using primers VP17 and VP18. (B) Amplification of the region containing putative intron using primers VP25 and MT3.

stop codons. The fact that the position of this region corresponded to that of an intron within the pea and *M. truncatula* Ogre sequences (NEUMANN *et al.* 2003 and our unpublished data), in addition to further computer analysis using NetGene2 server (HEBSGAARD *et al.* 1996), strongly suggested that this region could also represent an intron sequence. Moreover, the eventual removal of this predicted intron sequence by splicing would result in joining of ORF2 (+2 frame; 4403 bp) with ORF3 (+1 frame; 3464 bp) into one frame encoding a polyprotein of 2622 amino acids.

Transcriptional activity of *V. pannonica* Ogre elements was tested using RT–PCR with total RNA isolated from leaves, roots, and flowers. Primers were designed to amplify the ORF1 sequence as this region is specific for Ogre elements and is absent in all other types of retrotransposons described so far. The products of expected length were detected in all three organs tested (Figure 3A). To test the splicing of the predicted intron between ORF2 and ORF3, primers directed toward this region were also used. Although the corresponding RNA was again detected in all samples, the size of the amplified fragments did not correspond to the spliced sequence (Figure 3B). Cloning and sequence analysis of these fragments revealed 92–97% similarity to the corresponding region in Ogre-VP1. Comparison of the RT–PCR-amplified sequences with genomic sequences confirmed that none of them was spliced.

Abundance of Ogre-like sequences in Vicia and other legume species: The copy number of Ogre elements in the *V. pannonica* genome was measured using two different approaches. The first estimate was based on quantitative dot-blot hybridization of serial dilutions of genomic DNA and control fragments (cloned Ogre sequences) with probes derived from four regions of the element (see Figure 2A for the probe positions). Resulting signals corresponded to 1–5 $\times$ 10^5 copies/haploid genome (1C) for the LTR probe and to 1–2 $\times$ 10^5 copies/1C for each of the remaining probes (ORF1, ORF2, and ORF3). The second method employed hybridization of probes for LTR, ORF1, or ORF2 to a shotgun, short-insert library of *V. pannonica* genomic DNA. Of 6530 clones screened, 409, 169, or 237 clones, respectively, displayed positive hybridization signal. Considering the length of individual probes (1615, 1518, and 1999 bp) and using formulas described in MATERIALS AND METHODS, the estimated copy number per 1C was 2×10^5 for LTR, 8×10^4 for ORF1, and 9×10^4 for ORF2. Thus, the copy-number calculations based on these two principally different methods were in good accordance and a copy number of 1×10^5 copies/1C can be considered a reliable estimation. Taking into account the genome size of *V. pannonica* (6.75 pg; RAINA and REES 1983) and assuming that all element copies are full length, the copy number of 10^5/1C corresponds to 38% of the genome being made up by Ogre sequences.

To test if such a high amplification of Ogre elements resulted in their accumulation in specific genomic regions or if it was accompanied by an overall increase of the element copies throughout the entire genome, we performed detection of the Ogre repeats on mitotic chromosomes using *in situ* hybridization. These experiments showed that the elements were dispersed over the entire genome, spanning all of the chromosomes (Figure 4). The only exception to this homogenous chromosome labeling was the subtelomeric regions of the short chromosome arms, which produced weaker signals due to the presence of the highly amplified satellite repeat VicTR-A (MACAS *et al.* 2000).

The quantitative dot-blot hybridization was also used to estimate the abundance of Ogre-like sequences in 12 Vicia species and in several other legumes (Table 2). Two of these species, *V. melanops* and *V. hybrida*, were found to contain highly amplified Ogre elements, which were estimated to reach up to 1×10^5 and 1–5 $\times$ 10^4 copies/1C, respectively. These findings were in agreement with the results of Southern blot experiments in which only these two species showed prominent signals

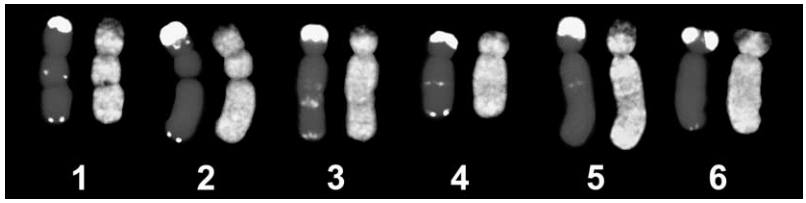


FIGURE 4.—FISH on metaphase chromosomes of *V. pannonica*. Two images of each chromosome type are shown. (Left of chromosome pair) DAPI-stained chromosomes (shaded) hybridized with the VicTR-A repeat (open signals). (Right of chromosome pair) The same chromosome hybridized with the probe specific for Ogre-VP1.

in addition to *V. pannonica* (Figure 5). Other Vicia species produced hybridization signals corresponding to lower numbers of Ogre sequences; however, the numbers estimated for individual species varied, depending on the probe used (Table 2). This observation most likely reflects sequence divergence of the corresponding regions (*e.g.*, ORF1) among species or eventual cross-hybridization of probes derived from regions conserved among various groups of retroelements (ORF3). This is even more evident for the estimates made for *P. sativum*, which were considerably lower than those made using Pisum-derived Ogre probes (NEUMANN *et al.* 2003 and Table 2). The species from other genera, including *Vigna unquiculata*, *Cicer arietinum*, *Glycine max*, *Lotus angustifolius*, *Phaseolus vulgaris*, and *M. truncatula*, produced no or very weak hybridization, not exceeding signals corresponding to <100–500 copies/1C (our unpublished data).

In addition to previously described Ogre sequences (NEUMANN *et al.* 2003), a number of partial clones of repetitive elements from *V. melanops*, *V. sativa*, and *V. narbonensis* were identified as most similar to Ogre-VP elements in BLAST and FASTA homology searches (Table 3). The high-sequence similarities strongly suggest that these clones correspond to various regions of Ogre-like elements, including ORF1.

DISCUSSION

A crucial role of the amplification and eventual elimination of retroelement sequences in plant genome evolution has become evident through the accumulation of sequence data and subsequent comparative analysis of species with different genome sizes. For example, a comparison of the *Adh1*-containing orthologous regions in maize and sorghum demonstrated that while >60% of this region in maize is composed of LTR retrotransposons, no retrotransposons were detected in the orthologous region in sorghum (SANMIGUEL *et al.* 1996; SANMIGUEL and BENNETZEN 1998). These LTR retrotransposons constitute 49–78% of the maize genome, suggesting that the threefold greater genome size of maize compared to sorghum was a result of retrotransposon amplification (SANMIGUEL and BENNETZEN 1998). Similar role of retroelements in genome-size evolution has also been revealed in other species (SHIRASU *et al.* 2000; WICKER *et al.* 2001, 2005). In general, all high-copy-number retrotransposons were identified in

TABLE 2

Estimated copy numbers of Ogre sequences per haploid genomes (1C) of selected Vicia species and *P. sativum*

Section	Species	1C (Mbp)	1C (pg)	Dot-blot probes				Other estimates	% in the genome[d]
				LTR	ORF1	ORF2	ORF3		
Hypechusa	*V. pannonica*	6,615	6.75	1–5 × 10^5	1 × 10^5	1 × 10^5	1 × 10^5	1 × 10^5 [a]	38
	V. hybrida	6,640	6.78	1 × 10^5	1 × 10^4	1 × 10^4	1–5 × 10^4	—	4
	V. lutea	7,252	7.40	0.5–1 × 10^4	0.5–1 × 10^3	5 × 10^2	1–5 × 10^3	—	0.2
	V. melanops	9,800	10.00	5 × 10^5	1 × 10^5	1 × 10^5	1 × 10^5	1 × 10^5 [b]	26 (26)[b]
Cracca	*V. villosa*	2,230	2.28	5 × 10^3	0.5–1 × 10^3	1 × 10^3	5 × 10^3	—	1
Attosa	*V. sepium*	4,582	4.68	1 × 10^4	1 × 10^3	5 × 10^3	1–5 × 10^4	—	3
Vicia	*V. sativa*	2,205	2.25	5 × 10^3	1–5 × 10^3	1–5 × 10^3	1–5 × 10^4	1 × 10^4 [b]	6 (12)[b]
	V. grandiflora	3,283	3.35	0.5–1 × 10^4	1 × 10^3	5 × 10^3	1–5 × 10^4	—	4
Wiggersia	*V. lathyroides*	2,573	2.63	1 × 10^2	0–50	0.5–1 × 10^2	5 × 10^3	—	0.1
Narbonensis	*V. narbonensis*	7,130	7.28	1–5 × 10^3	1 × 10^3	1–5 × 10^3	1–5 × 10^4	—	2
Faba	*V. faba*	13,255	13.50	1 × 10^3	0.5–1 × 10^2	1–5 × 10^2	1–5 × 10^4	—	0.1
Peregrina	*V. peregrina*	9,286	9.48	1–5 × 10^3	1–5 × 10^2	1 × 10^3	1–5 × 10^4	—	0.3
	V. michauxii	8,134	8.30	1 × 10^3	1 × 10^2	1 × 10^3	1 × 10^4	—	0.3
	P. sativum	4,337	4.43	1–5 × 10^2	5 × 10^2	1–5 × 10^3	1–5 × 10^4	1 × 10^4 [c]	3 (6)[c]

[a] Based on the number of positively hybridizing clones in the *V. pannonica* library.
[b] According to NOUZOVÁ *et al.* (2001).
[c] According to NEUMANN *et al.* (2003).
[d] Calculations were based on estimated copy numbers of the ORF2 probe. Values in parentheses show genome proportions calculated using estimates obtained with other species-specific Ogre probes (as shown in the column "Other estimates").

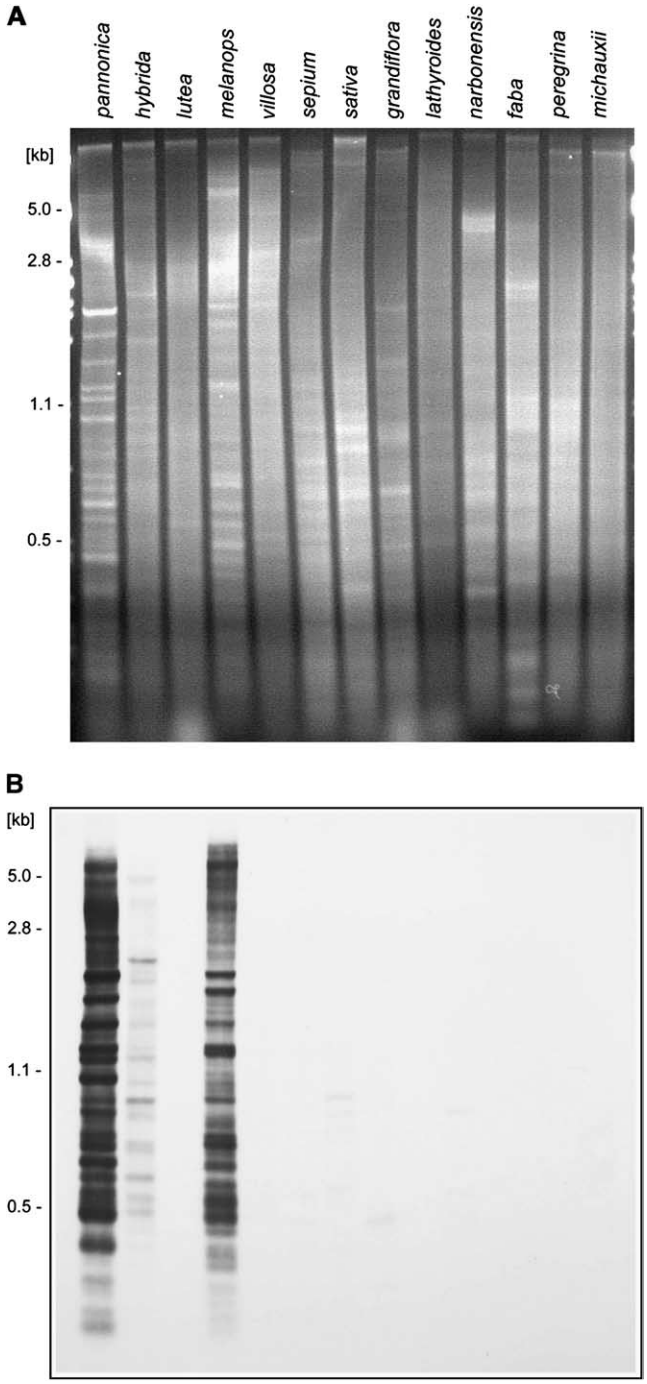


FIGURE 5.—Agarose gel electrophoresis (A) and corresponding Southern blot (B) of genomic DNAs of selected Vicia species digested with *Taq*I and probed with the whole VP-cosC6 insert sequence.

species with large genomes (PEARCE *et al.* 1996, 1997; MEYERS *et al.* 2001; MUNIZ *et al.* 2001; KENTNER *et al.* 2003) while species with small genomes contain these elements at relatively low copy numbers. The latter is the case of well-characterized genomes of *A. thaliana* (130 Mb) and *Oryza sativa* (430 Mb) in which retrotransposons make up only 4 and 22% of the genome (MA *et al.* 2004; PETERSON-BURCH *et al.* 2004; ZHANG and WESSLER 2004). All plant genomes characterized so far are composed of many diverse families of LTR retrotransposons with different degrees of amplification. The copy number of elements belonging to one family can vary greatly from a few to $\sim 10^5$ copies/genome (KUMAR and BENNETZEN 1999; FESCHOTTE *et al.* 2002). The most abundant plant LTR retrotransposons, such as PREM-2, Opie, or Huck-2 in *Zea mays* and IRRE in Iris, can make up to 10% of the genome (SANMIGUEL and BENNETZEN 1998; MEYERS *et al.* 2001; KENTNER *et al.* 2003; CHANTRET *et al.* 2005). Thus, concurrent amplification of elements belonging to several families can account for considerable increases in genome size as demonstrated for Zea species (SANMIGUEL and BENNETZEN 1998; MEYERS *et al.* 2001). Our results described here demonstrate that even the amplification of a single retroelement family can increase the genome size by >50% (based on the calculation that this element makes up 38% of the *V. pannonica* genome). Although the Ogre sequences were detected in all Vicia species tested, their abundance differed by several orders of magnitude (Table 2). Whereas the copy numbers in some species could be underestimated due to sequence divergence of the elements, this great variability in copy numbers was evident even among closely related species, each belonging to the taxonomic section Hypechusa. Among the four species tested, the Ogre sequences were ~10-fold less abundant in *V. lutea* than in *V. hybrida* and 100-fold less abundant than in *V. pannonica* and *V. melanops*. Interestingly, the genome size of *V. lutea* is similar to those of the other three species, suggesting that other repetitive elements may have been amplified there instead of Ogre. The amplification of other elements probably accompanied the expansion of the genome size in *V. melanops*, as the copy number of the Ogre family in this species is the same as in *V. pannonica*, but its genome is larger (Table 2). Therefore, although it is evident from the differences in Ogre copy numbers in individual species that these elements have played a crucial role in the increase of genome size for at least some Vicia species, there is no simple correlation between abundance of Ogre sequences and the genome size in this genus.

The differences in copy numbers observed among Vicia species could be in part explained by a loss of Ogre sequences in some genomes due to unequal homologous recombination and illegitimate recombination, which were shown to be responsible for deletions of retrotransposon sequences from genomes in several plant species (SHIRASU *et al.* 2000; DEVOS *et al.* 2002; VITTE and PANAUD 2003; WICKER *et al.* 2003; MA *et al.* 2004). An indicator of unequal recombination is a presence of solo LTRs, which remain in a genome after a recombination event. The copy number estimated for LTR sequences in several species was considerably higher compared to inside regions (*e.g.*, *V. lutea* and *V. hybrida*, Table 2), although the LTR sequences are believed to be

TABLE 3

Ogre-like sequences identified in Vicia species by FASTA sequence similarity searches using Ogre-VP1 as a query

Species	Accession no.	Similarity (%/length)	*E*-value	Region	Copy nos. (copies/1C)[a]
V. melanops	AJ391781	88/666	1.6×10^{-70}	LTR	10^5–10^6
	AJ391800	75/982	1.7×10^{-86}	LTR	10^4–10^5
	AJ391797	78/493	1.1×10^{-37}	ORF1/spacer	10^5–10^6
	AJ391792	80/479	1.2×10^{-94}	ORF2	10^5–10^6
	AJ391793	86/222	2.6×10^{-46}	ORF2	10^4–10^5
	AJ391794	90/337	4.5×10^{-84}	ORF2	10^4–10^5
	AJ391802	75/211	2.2×10^{-30}	ORF2	10^4–10^5
	AJ391799	85/332	6.9×10^{-62}	ORF2/intron	10^5–10^6
	AJ391754, AJ391755, AJ391759, AJ391760, AJ391761, AJ391764, AJ391765	87/490–92/757	2.3×10^{-105}–4.1×10^{-187}	ORF3	Unknown
	AJ391774	91/308	7.3×10^{-49}	ORF3/3′-UTR	Unknown
V. sativa	AJ391782	69/781	4.8×10^{-30}	LTR	10^4–10^5
	AJ391804	64/118	8.1	5′-UTR	10^4
	AJ391758, AJ391762, AJ391763, AJ391766, AJ391767, AJ391769	68/275–87/421	9.1×10^{-27}–1.4×10^{-88}	ORF3	Unknown
	AJ391776, AJ391778	82/65, 85/65	6×10^{-6}, 7.8×10^{-5}	3′-UTR	Unknown
V. narbonensis	AJ391771	62/320	1.9×10^{-13}	ORF1	Unknown
	AJ391756, AJ391757, AJ391768, AJ391770	70/382–68/664	7.3×10^{-71}	ORF3	Unknown
	AJ391777	76/107	0.0029	3′-UTR	Unknown

[a] Copy numbers are according to Nouzová *et al.* (2001).

less conserved and thus less likely to hybridize with a probe from another species than the internal coding sequences. Thus, it is possible that in these species the reduction of the genome size through the elimination of Ogre sequences took place, leaving an excess of solo LTRs.

Molecular mechanisms facilitating the eventual regulation of retroelement amplification in plant genomes are only poorly understood. Mechanisms that are likely to play a role in the suppression of mobile elements include transcriptional silencing through DNA methylation (Hirochika *et al.* 2000; Wright and Voytas 2002; Liu *et al.* 2004), chromatin modifications (Jackson *et al.* 2002), and post-transcriptional silencing by RNA interference mechanism (Timmons 2002). The high amplification of Ogre in a limited number of species and its occurrence in only moderate copy numbers in others can be explained either by an insufficient function of mechanisms suppressing transposition of retroelements or by the escape of Ogre elements from such mechanisms. Since virtually all prominent bands visible on digested genomic DNA of *V. pannonica* hybridized to Ogre, it seems that only this family of retrotransposons was amplified to high copy numbers, whereas other retroelements remained suppressed. Considering the high sequence similarity among Ogre sequences isolated from *V. pannonica*, the high level of conservation in restriction sites, and the recent insertion of Ogre-VP1 element (inferred from the similarity of its LTRs), it is likely that the burst of amplification of the Ogre family happened quite recently. As these elements are still transcribed in *V. pannonica*, the amplification of the Ogre family in this species may not be over yet. In this respect, it is interesting to mention that the transcription of Ogre elements also was detected in *P. sativum* (Neumann *et al.* 2003) containing one order of magnitude fewer copies compared to *V. pannonica* and in *M. truncatula* having a small genome (0.48 pg or 466 Mbp; Arumuganathan and Earle 1991), which, consequently, can be occupied by relatively few Ogre elements. This could imply that transcription itself is not sufficient for transpositional activity and that post-transcriptional regulation plays an important role in the suppression of transposition.

From the results described here, as well as from our previous findings (Neumann *et al.* 2003), it is evident that Ogre elements represent a group of retrotransposons capable of inducing significant changes in the genome size of some plant species. Up to now, full-length Ogre elements have been detected in only a relatively narrow range of legume taxa, including the genera of Pisum, Medicago, and Vicia. However, except

for the hybridization-based detection described here no attempts were performed to identify related elements in other plants. Moreover, no or very weak hybridization signals obtained in species from more distant genera (Vigna, Cicer, Glycine, Lotus, and Phaseolus) do not necessarily indicate the absence of Ogre elements but could rather be caused by their further sequence divergence. Thus, the alternative approaches on the basis of sequence analysis of regions typical for the Ogre family, including mainly ORF1 and pbs, should be employed to find Ogre-like elements in other species. Future experiments should also be directed to structural and functional analysis of the extremely long Ogre LTRs harboring regulatory regions controlling the element replication cycle, which could provide clues for explaining the high amplification rate of Ogre elements in some species.

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